Body Type Quiz Design Specification

Overview

The LYKE Body Type Quiz is a 6-question assessment that determines a user's body shape, stature, and build. The quiz uses a weighted scoring algorithm to classify users into one of five body shapes, combined with stature and build modifiers. Results are used to match users with creators who have similar body profiles.

Design Goals

1. **Accessibility** — Anonymous access allows social media sharing and viral engagement
2. **Accuracy** — Weighted scoring matrices provide nuanced classification
3. **Editability** — Results auto-populate but users can override selections
4. **Privacy** — No personal measurements required; uses relative proportions only

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Body Shape Classifications

The quiz determines one of five body shapes based on proportional characteristics:

ID	Shape	Description	Key Characteristics
1	**Hourglass**	Balanced bust and hips with defined waist	Shoulders and hips approximately equal width; clearly defined waistline
2	**Pear**	Hips wider than shoulders	Weight carried in lower body; narrower shoulders; defined waist
3	**Apple**	Fuller midsection with slimmer legs	Weight carried in midsection; less waist definition; slimmer lower body
4	**Rectangle**	Balanced proportions, less waist definition	Shoulders, waist, and hips similar width; athletic or straight silhouette
5	**Inverted Triangle**	Shoulders wider than hips	Broader shoulders/bust; narrower hips; weight in upper body

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Stature and Build Modifiers

In addition to body shape, the quiz captures two modifiers:

Stature (Question 5)

Value	Enum	Description
Petite	`Stature.Petite` (1)	Shorter stature, typically under 5'3" / 160cm
Average	`Stature.Average` (2)	Medium height range
Tall	`Stature.Tall` (3)	Taller stature, typically over 5'7" / 170cm

Build (Question 6)

Value	Enum	Description
Standard	`Build.Standard` (1)	Standard clothing size range
Plus	`Build.Plus` (2)	Plus-size clothing range

Quiz Questions

Question 1: Shoulder-Hip Comparison

Prompt: "When you look in the mirror, how do your shoulders compare to your hips?"

Index	Answer	Indicates
0	My shoulders are noticeably wider than my hips	Inverted Triangle, Apple, Rectangle
1	My shoulders and hips are about the same width	Hourglass, Rectangle
2	My hips are noticeably wider than my shoulders	Pear

Question 2: Waist Definition

Prompt: "How would you describe your natural waistline?"

Index	Answer	Indicates
0	Very defined — my waist is noticeably narrower than my bust and hips	Hourglass, Pear
1	Moderately defined — there's some curve but not dramatic	Neutral (all shapes)
2	Minimal definition — my waist, bust, and hips are similar measurements	Rectangle, Apple

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Question 3: Weight Distribution

Prompt: "When you gain weight, where does it tend to go first?"

Index	Answer	Indicates
0	Upper body (shoulders, arms, bust area)	Inverted Triangle
1	Midsection (stomach, waist, back)	Apple
2	Lower body (hips, thighs, bottom)	Pear
3	Evenly distributed throughout my body	Hourglass, Rectangle

Question 4: Clothing Fit Challenges

Prompt: "When shopping for clothes, where do you most often need adjustments?"

Index	Answer	Indicates
0	Shoulders/bust area is often too tight	Inverted Triangle
1	Waist/midsection needs more room	Apple
2	Hips/thighs are snug while waist gaps	Pear, Hourglass
3	Clothes fit similarly throughout — alterations are consistent	Rectangle

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Question 5: Stature

Prompt: "How would you describe your height?"

Index	Answer	Maps To
0	Petite (under 5'3" / 160cm)	`Stature.Petite`
1	Average (5'3" - 5'7" / 160-170cm)	`Stature.Average`
2	Tall (over 5'7" / 170cm)	`Stature.Tall`

Question 6: Build

Prompt: "Which best describes your typical clothing size range?"

Index	Answer	Maps To
0	Standard sizes (XS-XL, 0-14)	`Build.Standard`
1	Plus sizes (1X+, 16+)	`Build.Plus`

Scoring Algorithm

Scoring Matrices

Questions 1-4 use weighted scoring matrices. Each answer awards points to multiple body shapes simultaneously. The shape with the highest total score wins.

Matrix Format: `[Hourglass, Pear, Apple, Rectangle, Inverted Triangle]`

Q1: Shoulder-Hip Comparison

```
Answer 0 (Shoulders wider):    [0, 0, 1, 1, 3]
Answer 1 (Same width):         [2, 1, 1, 2, 0]
Answer 2 (Hips wider):         [0, 3, 0, 1, 0]
```

Q2: Waist Definition

```
Answer 0 (Very defined):       [3, 2, 0, 0, 1]
Answer 1 (Moderate):           [1, 1, 1, 1, 1]
Answer 2 (Minimal):            [0, 0, 2, 3, 1]
```

Q3: Weight Distribution

```
Answer 0 (Upper body):         [0, 0, 1, 0, 3]
Answer 1 (Midsection):         [0, 0, 3, 1, 0]
Answer 2 (Lower body):         [1, 3, 0, 0, 0]
Answer 3 (Evenly):             [2, 1, 1, 2, 1]
```

Q4: Clothing Fit Challenges

```
Answer 0 (Tight shoulders):    [0, 0, 0, 0, 2]
Answer 1 (Tight middle):       [0, 0, 2, 0, 0]
Answer 2 (Tight hips):         [1, 2, 0, 0, 0]
Answer 3 (Same everywhere):    [0, 0, 0, 2, 0]
```

Score Calculation

1. Initialize score array: `[0, 0, 0, 0, 0]` for each body shape
2. For each answer to Q1-Q4, add the corresponding row from the scoring matrix
3. The body shape with the highest total score is selected

Example:

```
User answers: Q1=1, Q2=0, Q3=3, Q4=2

Q1 (Same width):      [2, 1, 1, 2, 0]
Q2 (Very defined):    [3, 2, 0, 0, 1]
Q3 (Evenly):          [2, 1, 1, 2, 1]
Q4 (Tight hips):      [1, 2, 0, 0, 0]
─────────────────────────────────────
Total:                [8, 6, 2, 4, 2]

Winner: Hourglass (score: 8)
```

Maximum Possible Scores

Shape	Max Score	Achieved By
Hourglass	8	Q1=1, Q2=0, Q3=3, Q4=2
Pear	10	Q1=2, Q2=0, Q3=2, Q4=2

Shape	Max Score	Achieved By
Apple	8	Q1=0, Q2=2, Q3=1, Q4=1
Rectangle	8	Q1=1, Q2=2, Q3=3, Q4=3
Inverted Triangle	9	Q1=0, Q2=2, Q3=0, Q4=0

Tie-Breaker Logic

When two or more body shapes have equal highest scores, the tie-breaker priority determines the winner:

Priority Order (highest to lowest):

1. Hourglass
2. Pear
3. Rectangle
4. Apple
5. Inverted Triangle

Rationale: Hourglass and Pear are the most common body shapes in the general population, so ties are resolved toward these more prevalent classifications. This reduces edge cases where users might feel misclassified.

Result Label Formatting

The result label combines stature, build, and body shape into a human-readable string.

Formatting Rules

1. **Stature** — Include only if NOT "Average" (Average is the default/baseline)
2. **Build** — Include only if NOT "Standard" (Standard is the default/baseline)
3. **Shape** — Always included

Examples

Stature	Build	Shape	Result Label
Petite	Plus	Hourglass	"Petite Plus Hourglass"
Tall	Standard	Pear	"Tall Pear"
Average	Plus	Rectangle	"Plus Rectangle"
Petite	Standard	Apple	"Petite Apple"
Average	Standard	Inverted Triangle	"Inverted Triangle"

API Specification

Endpoint

```
POST /api/quiz/v1/calculate
```

Authentication: None required (anonymous access for shareability)

Request

```json
{
  "answers": [
    { "questionId": 1, "answerIndex": 1 },
    { "questionId": 2, "answerIndex": 0 },
    { "questionId": 3, "answerIndex": 3 },
    { "questionId": 4, "answerIndex": 2 },
    { "questionId": 5, "answerIndex": 0 },
    { "questionId": 6, "answerIndex": 1 }
  ]
}
```

Field	Type	Required	Description
`answers`	array	Yes	List of quiz answers
`answers[].questionId`	int	Yes	Question ID (1-6)
`answers[].answerIndex`	int	Yes	Zero-based answer index

Response

```json
{
  "success": true,
  "data": {
    "bodyTypeId": 1,
    "bodyTypeName": "Hourglass",
    "stature": "Petite",
    "build": "Plus",
    "resultLabel": "Petite Plus Hourglass",
    "resultDescription": "We'll show you content from creators with similar proportions."
  }
}
```

Field	Type	Description
`bodyTypeId`	int	Database ID of the body type (1-5)
`bodyTypeName`	string	Human-readable body shape name
`stature`	string	Stature classification (Petite, Average, Tall)

Field	Type	Description
`build`	string	Build classification (Standard, Plus)
`resultLabel`	string	Formatted display label
`resultDescription`	string	Explanation text for the user

Validation Rules

- `answers` array must not be empty
- `answers` array cannot exceed 10 items
- `questionId` must be between 1 and 10
- `answerIndex` must be >= 0 and <= 10

Error Response

```
{
  "success": false,
  "error": {
    "code": "INVALID_REQUEST",
    "message": "Quiz answers are required"
  }
}
```

Default Behaviors

Scenario	Behavior
Q5 (Stature) not answered	Defaults to `Average`
Q6 (Build) not answered	Defaults to `Standard`
Invalid question ID	Answer ignored, no error
Invalid answer index	Answer ignored, no error
All Q1-Q4 skipped	Returns shape with highest tie-breaker priority (Hourglass)
Empty answers array	Returns defaults: Hourglass, Average, Standard

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Database Integration

Body Type Mapping

Shape Name	Database ID
Hourglass	1
Pear	2

Shape Name	Database ID
Apple	3
Rectangle	4
Inverted Triangle	5

Body Profile Fields

Quiz results can be saved to the user's `BodyProfile` entity:

```
public class BodyProfile
{
    public int BodyTypeId { get; set; }       // From quiz result
    public Stature? Stature { get; set; }     // From Q5
    public Build? Build { get; set; }         // From Q6
    // ... other fields
}
```

Frontend Integration Notes

Suggested UI Flow

1. **Introduction screen** — Explain what the quiz determines and how results are used
2. **Questions 1-4** — Body shape assessment (progress indicator: 1/6 to 4/6)
3. **Question 5** — Stature selection (progress: 5/6)
4. **Question 6** — Build selection (progress: 6/6)
5. **Results screen** — Display result label with body shape illustration
6. **Confirmation** — Allow user to accept or manually adjust selections

Social Sharing

The anonymous endpoint enables sharing quiz results on social media:

- Generate shareable image/card with result label and silhouette
- Deep link back to LYKE app/website
- Track viral coefficient through referral codes (future enhancement)

Accessibility Considerations

- All questions should have clear, jargon-free language
- Include visual aids (silhouette illustrations) alongside text options
- Support screen readers with proper ARIA labels
- Allow keyboard navigation through answer options

Future Enhancements

1. **Confidence scoring** — Show users how strongly they match their result vs. alternatives
2. **Secondary type** — Display "You're also close to Pear" when scores are near-tied
3. **Retake flow** — Allow users to retake quiz and compare results
4. **A/B testing** — Test alternative question wordings for accuracy improvement
5. **Machine learning** — Train model on user feedback to improve scoring weights